

```
def plot_phase_attractor(data):  
    for trajectory in trajectories:  
        plt.scatter(trajectory.x, trajectory.y, c='magenta', s=0.1)
```

```
class StateSpaceRender(engine):  
    self.resolution = (3840, 2160)  
    self.channels = RGB  
    self.mapping = FLUID_DYNAMIC
```

```
if RENDER_MODE == DETERMINISTIC:  
    optimize_manifold()
```

```
class StateSpaceRend(data):  
    plt.tocestesot()
```

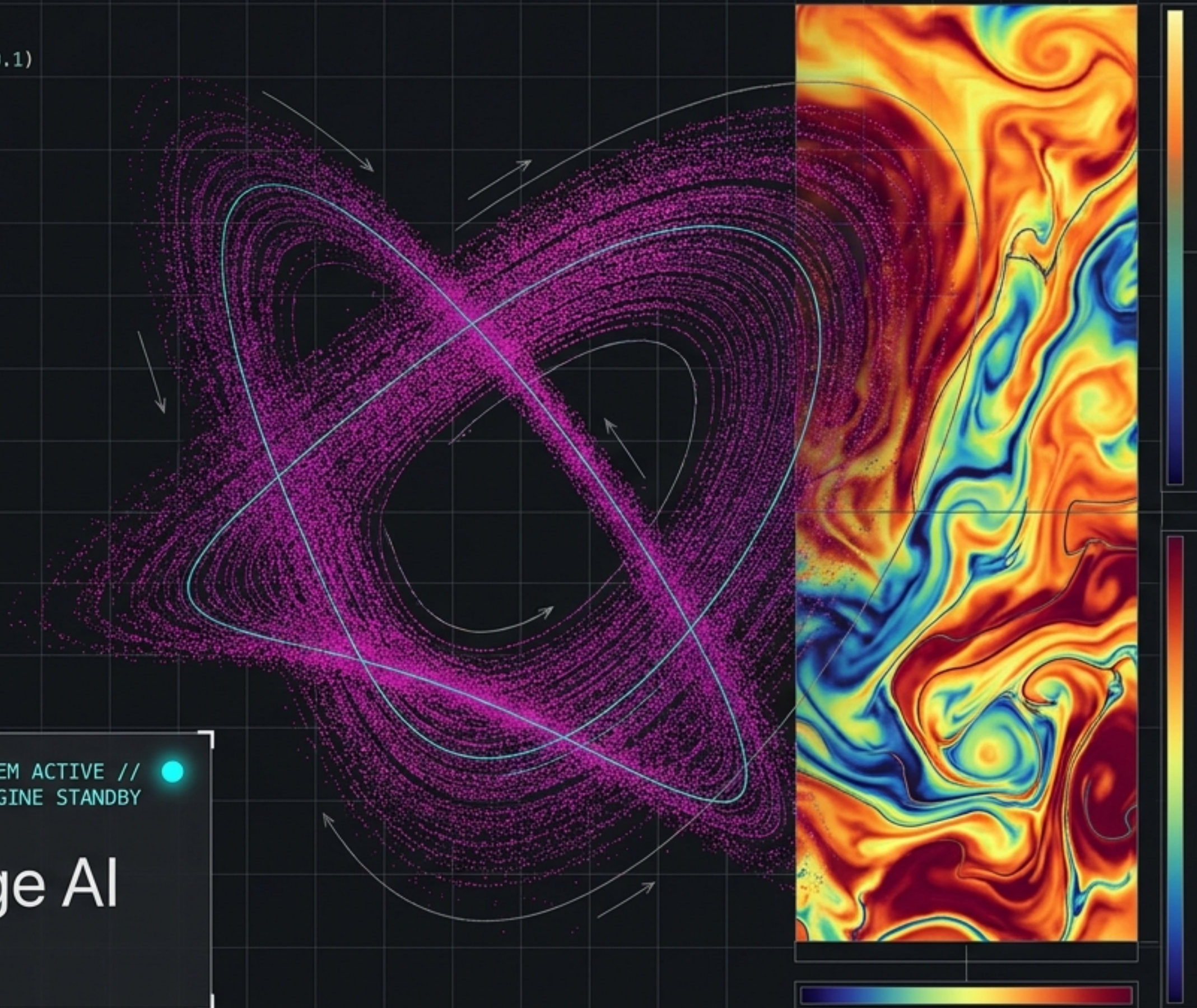
```
if RENDER_MODE == DETERMINISTIC:  
    optimize_mondel()
```

```
HASH: Bx4D5F5250_9B7E1A2C_F3A1B8D2  
LOG: System initialization complete.  
Telemetry streaming.  
VRAM allocation: 12.8GB.
```

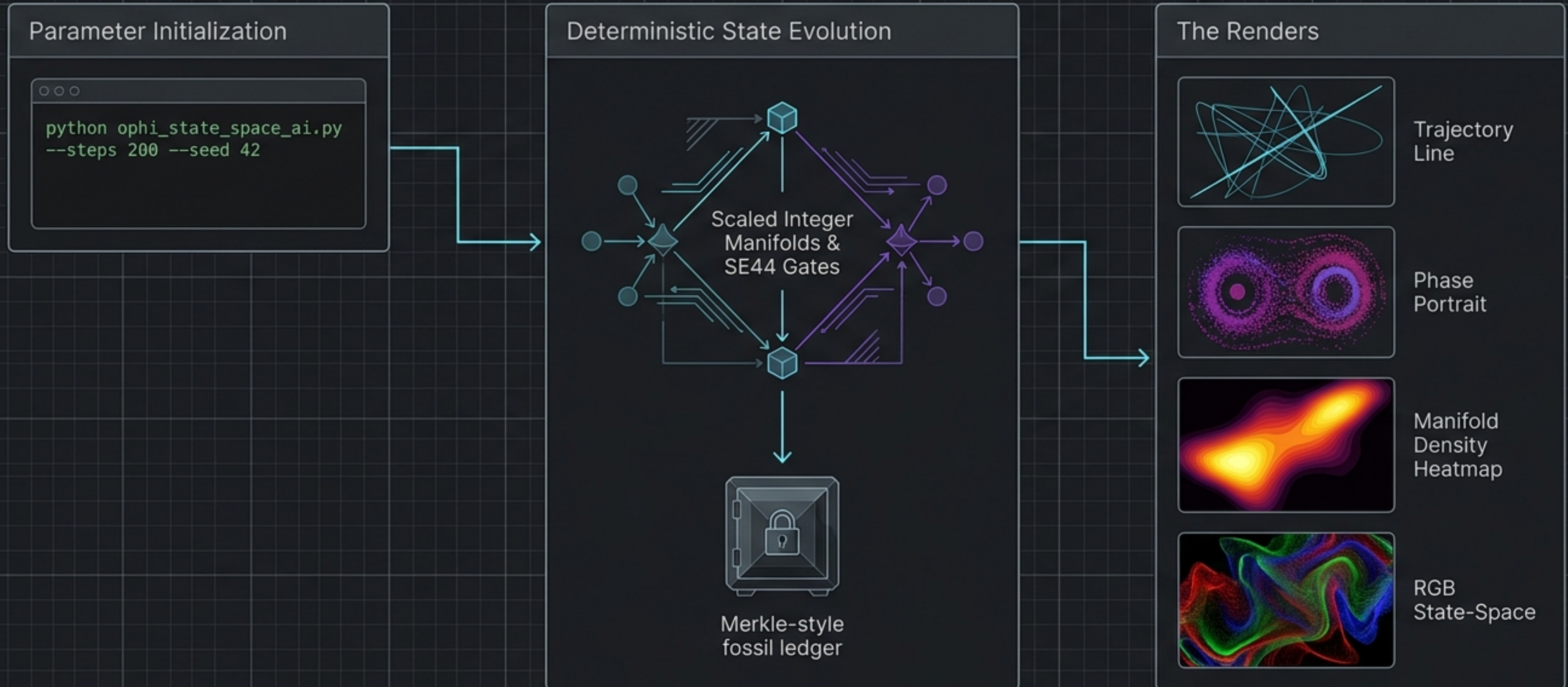
SYSTEM ACTIVE // 
RENDER ENGINE STANDBY

OPHI State Space Image AI

Deterministic Manifold Visualization

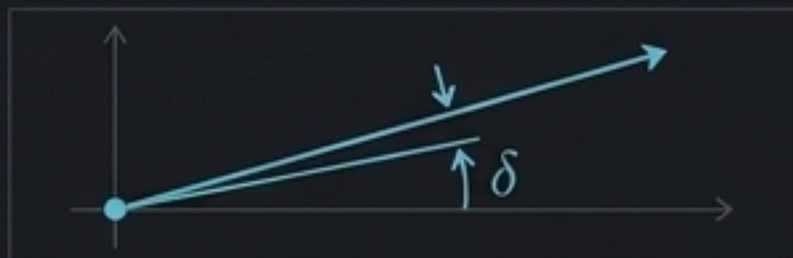


Pipeline Architecture: Code to Canvas



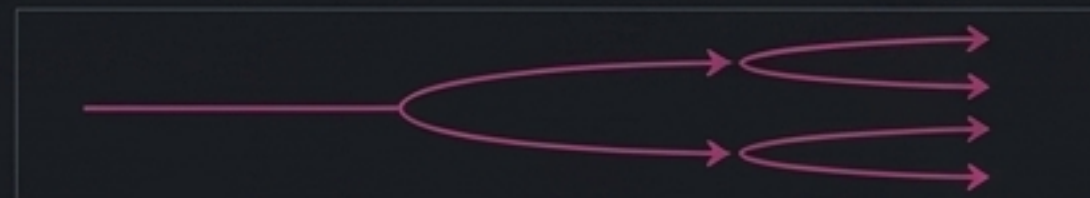
The Initial Seed

Determines the base trajectory.



The Chaotic Multiplier

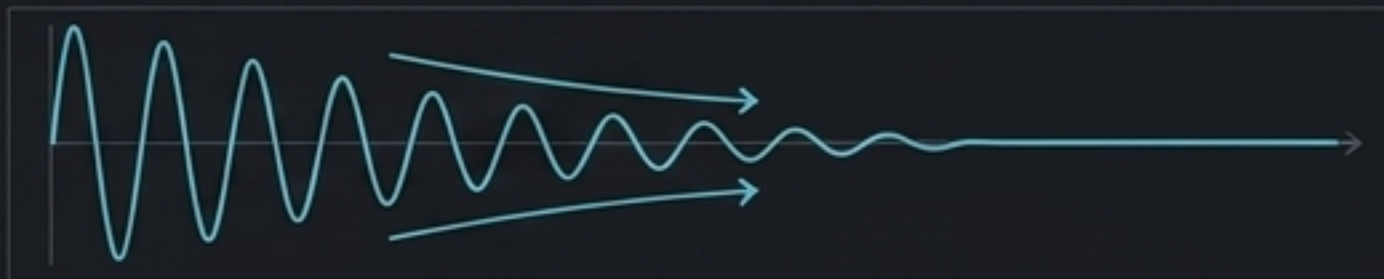
Typically set just above 1 (e.g., 1.001) to drive continuous systemic evolution.



$$\Omega = (\text{state} + \text{bias}) \times \alpha \times \text{reliability} \times \text{grounding}$$

The Systemic Dampeners

Controls visual sprawl and anchors the generated manifold.

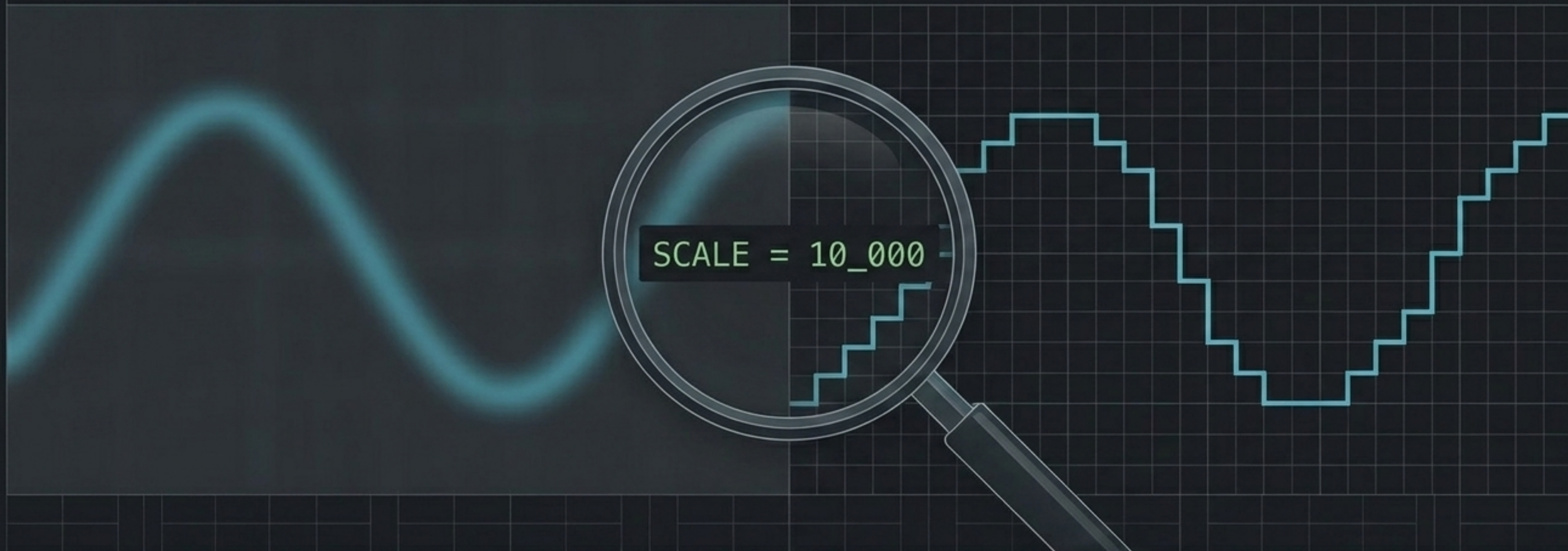


Floating-Point Drift

Microscopic CPU discrepancies break reproducibility.

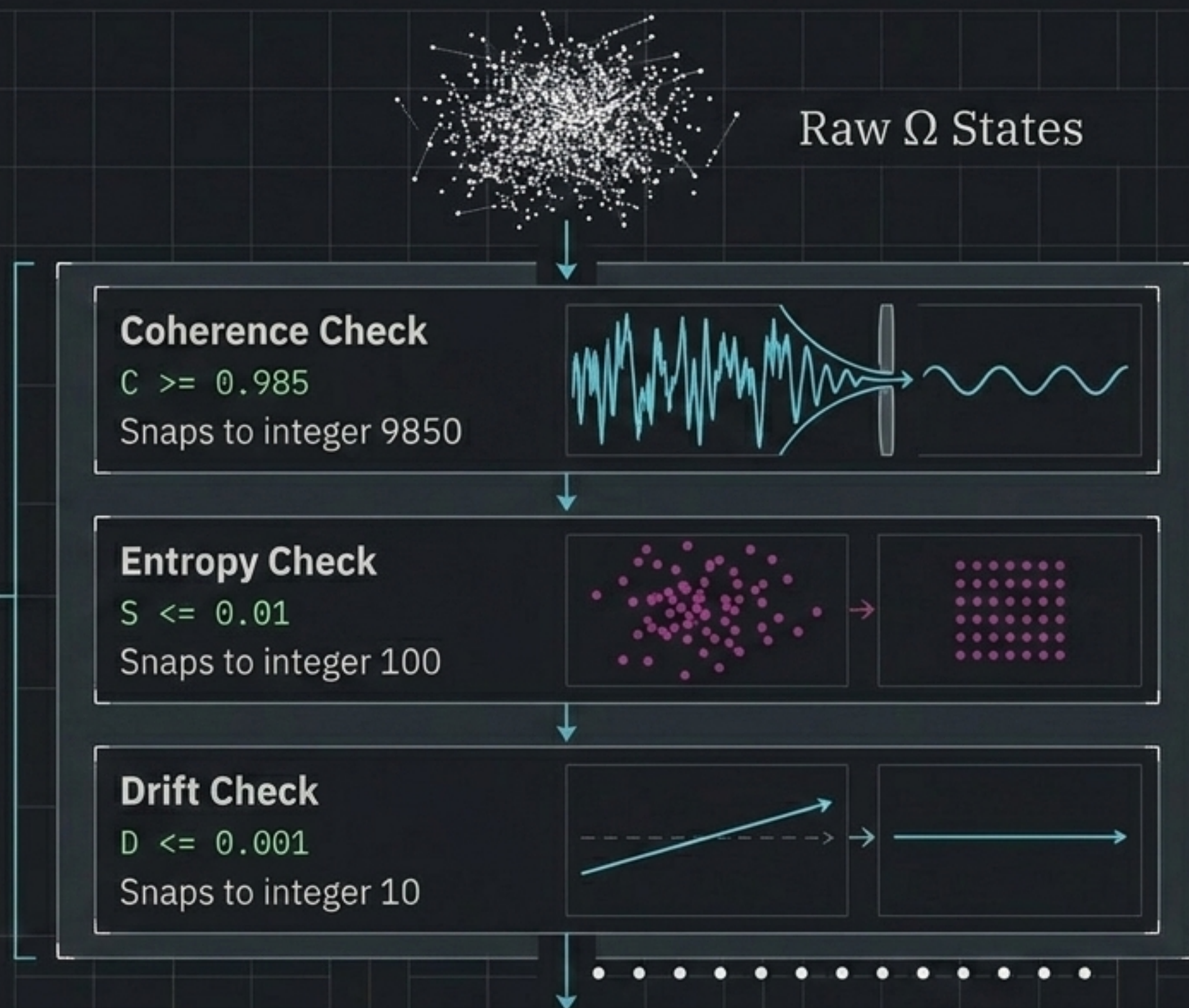
Scaled Integer Manifold

Continuous math converted into deterministic integer space.



The Result: Multiplying thresholds by 10,000 eliminates CPU-level float variations. Every algorithmic render is mathematically verifiable and pixel-perfect across all machines.

Bifurcation Diagram



Only states surviving this strict thermodynamic filtration influence the final aesthetic rendering queue.

State Bifurcation: The Consequence of the SE44 Gate

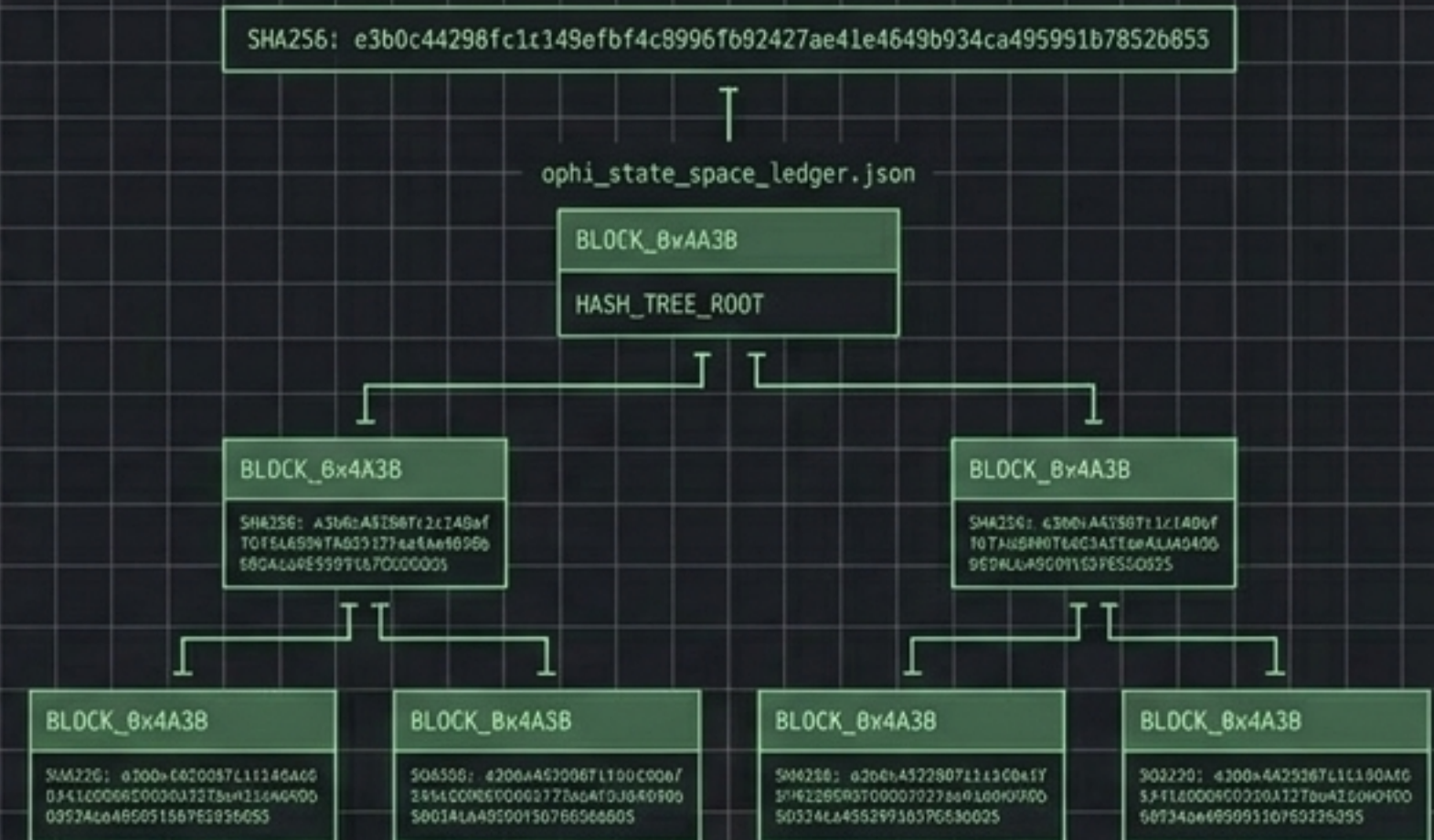
Mutable Shell



Rejected States

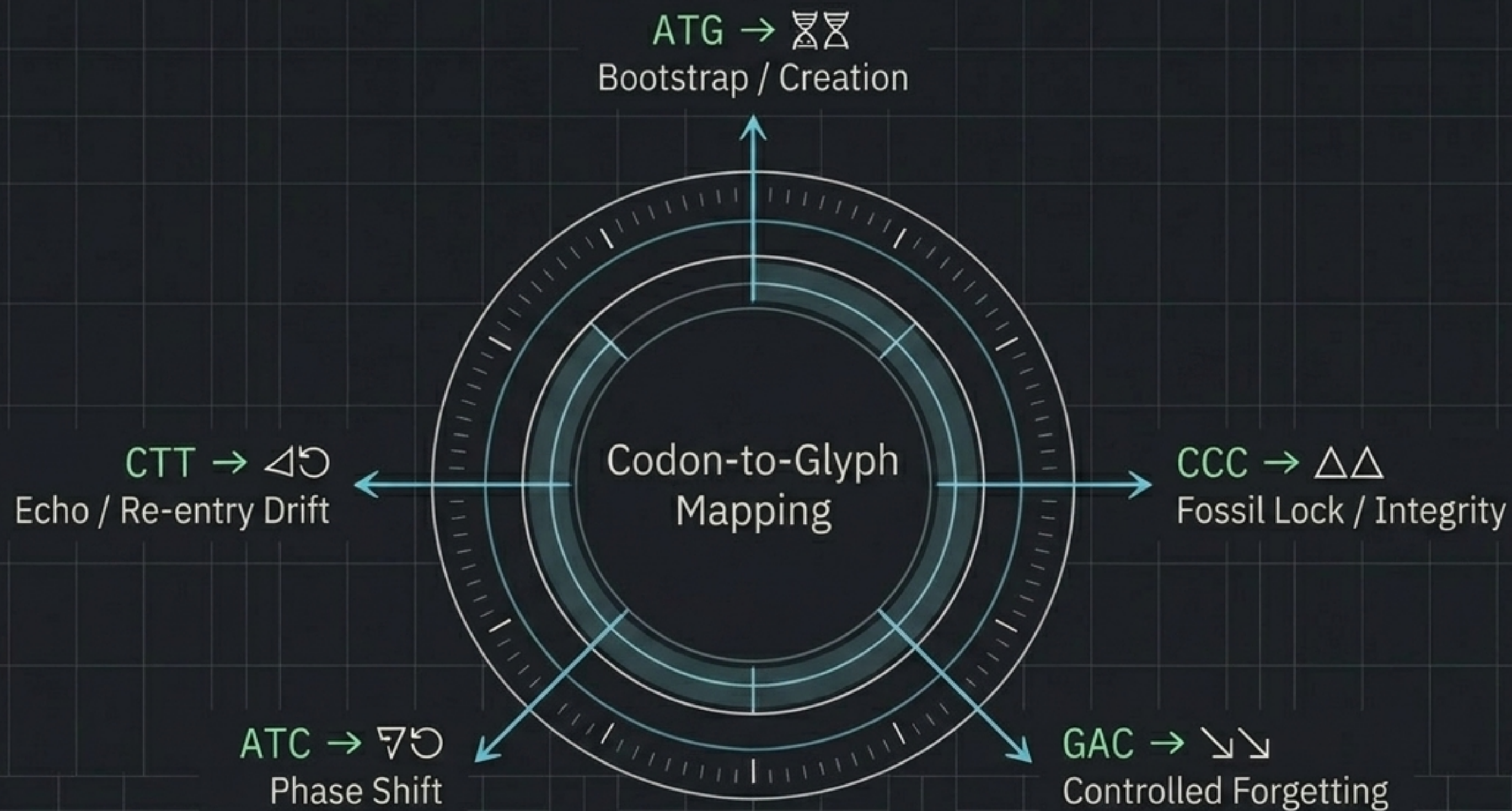
Discarded into a mutable shell. This isolation prevents visual noise and strictly preserves the mathematical integrity of the core aesthetic manifold.

Merkle Ledger



Accepted Fossils

Verified states are hashed and locked into an immutable JSON ledger. The resulting visual output is cryptographically tied directly to its mathematical history.



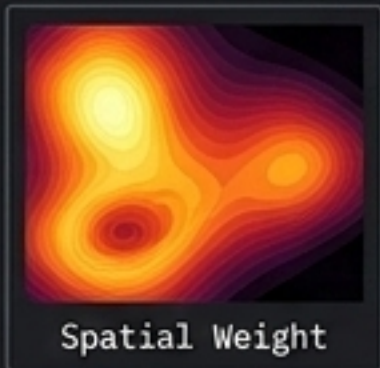
These triads act as systemic DNA, embedding translated metadata directly into the architecture of the state space.



File: 01_trajectory.png

Dimension: 1D Time-series

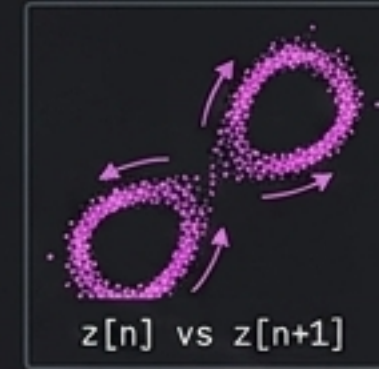
Function: Tracks Ω evolution over accepted step counts.



File: 03_density_heatmap.png

Dimension: 2D/3D Spatial Weight

Function: Accumulation of states in thermal probability buckets.



File: 02_phase_portrait.png

Dimension: 2D Temporal Geometry

Function: Maps current state $z[n]$ directly against future state $z[n+1]$.



File: 04_rgb_state_space_image.png

Dimension: Multidimensional Synthesis

Function: Full-color manifold rendering mapped to RGB channels.

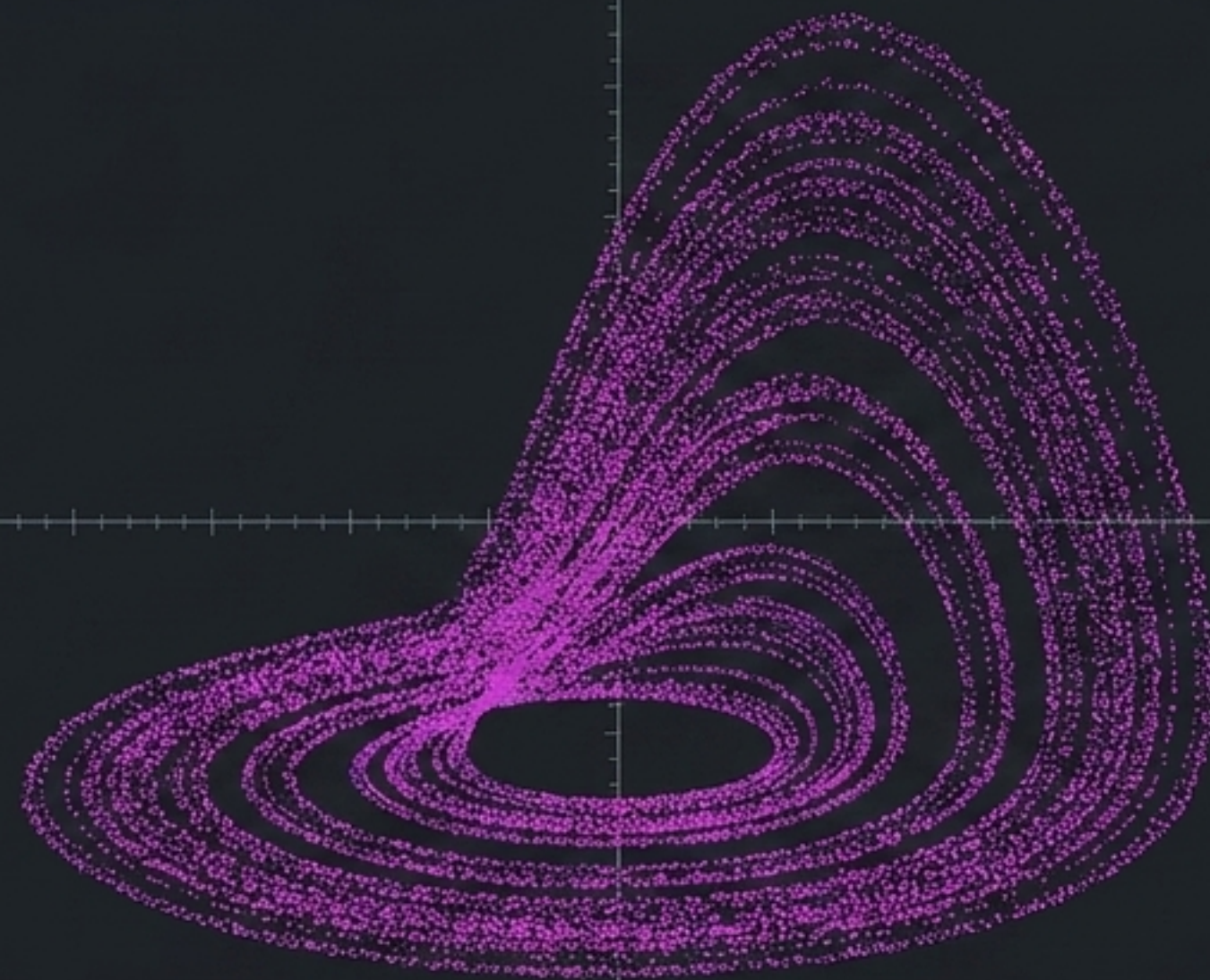
Canvas 1: Trajectory Line

High-frequency jaggedness
flags systemic drift.

Smooth curves indicate
locked coherence.

Visualizing the algorithmic heartbeat.
This linear mapping tracks the raw Ω evolution step-by-step.

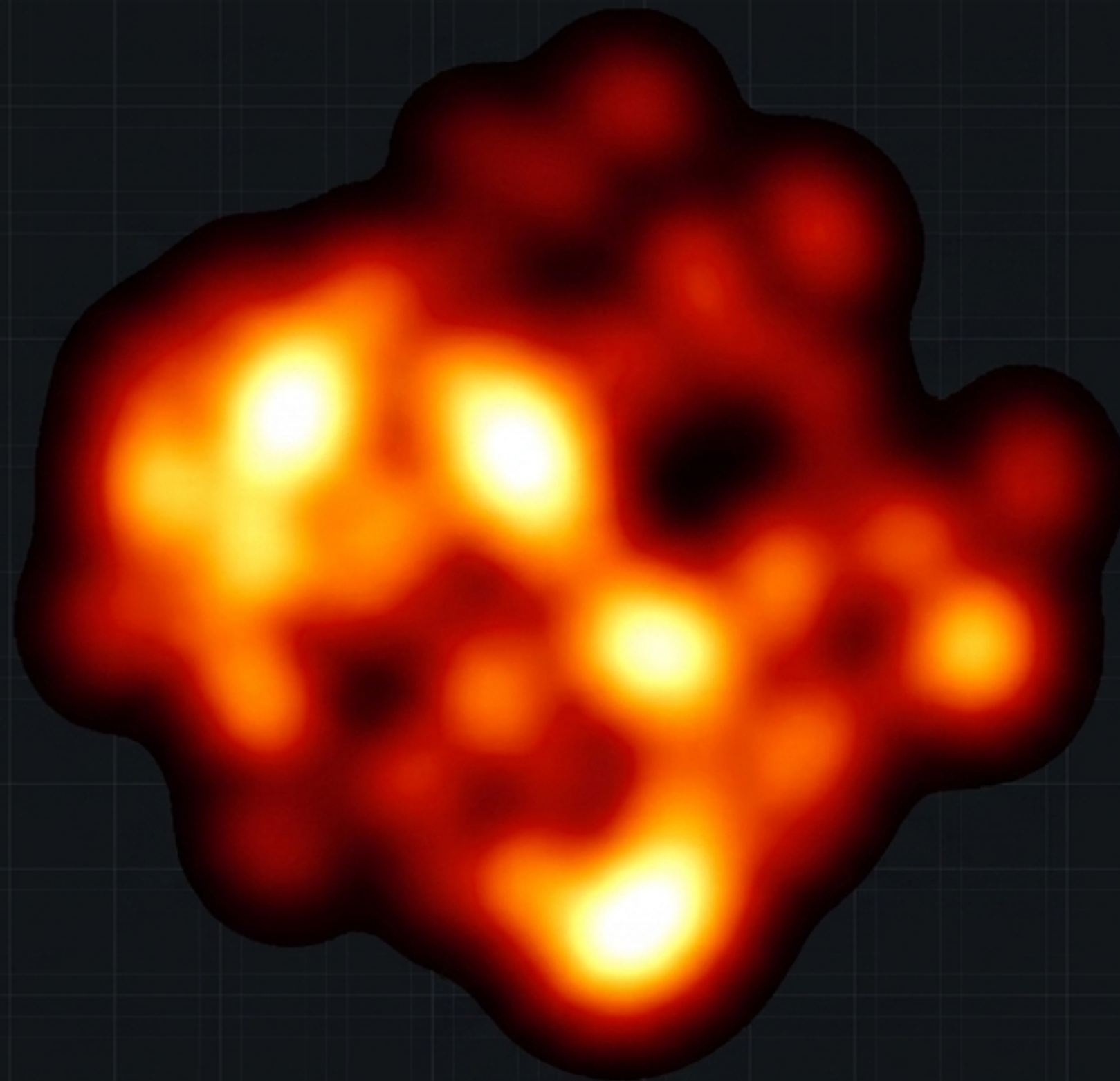
Canvas 2: Phase Portrait



Visualizing hidden geometries.

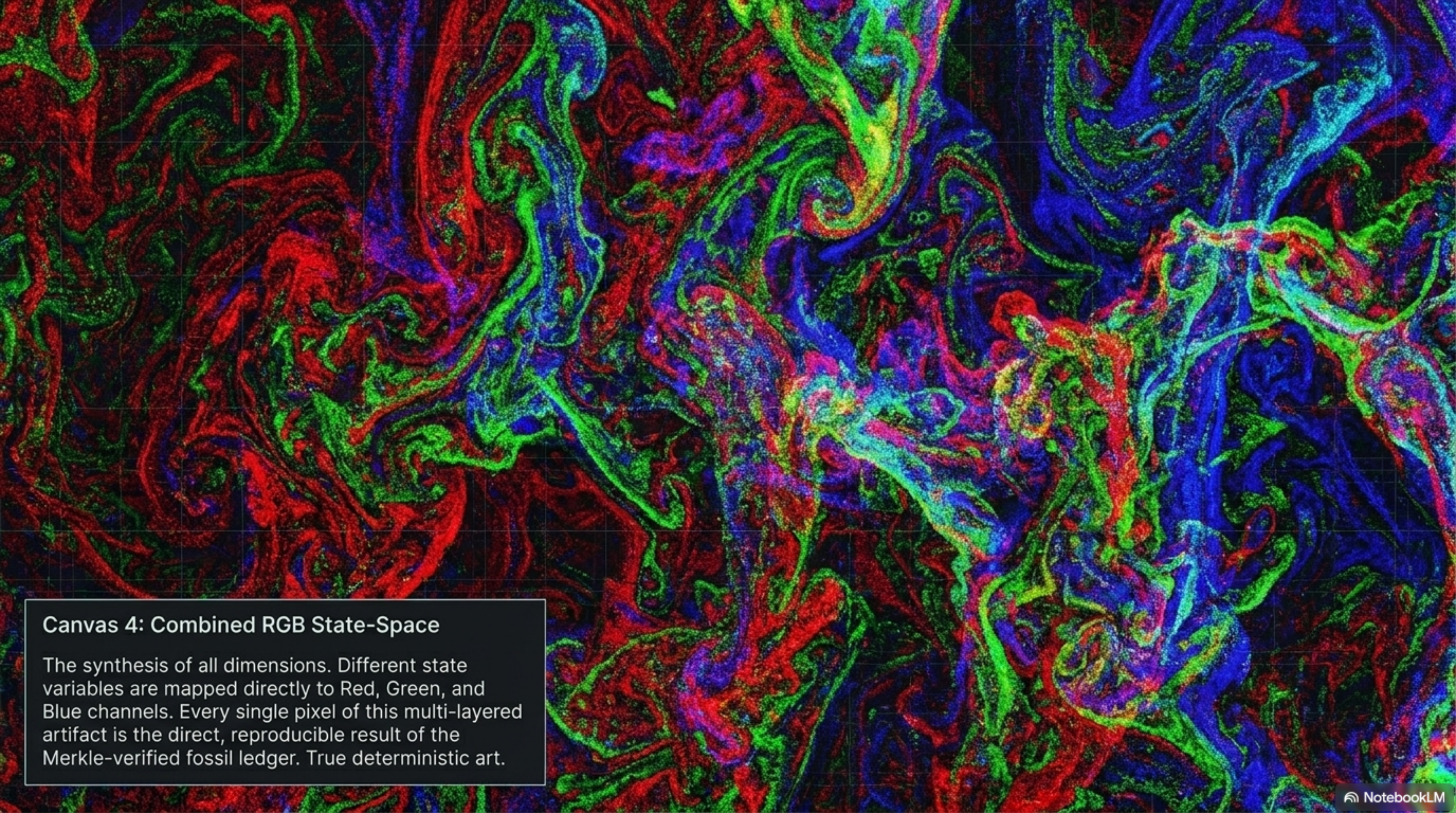
By mapping the current mathematical state $z[n]$ against the immediate future state $z[n+1]$, temporal data collapses into recognizable spatial structures, revealing the strict boundaries of the scaled integer manifold.

Canvas 3: Density Heatmap



Visualizing weight and probability.

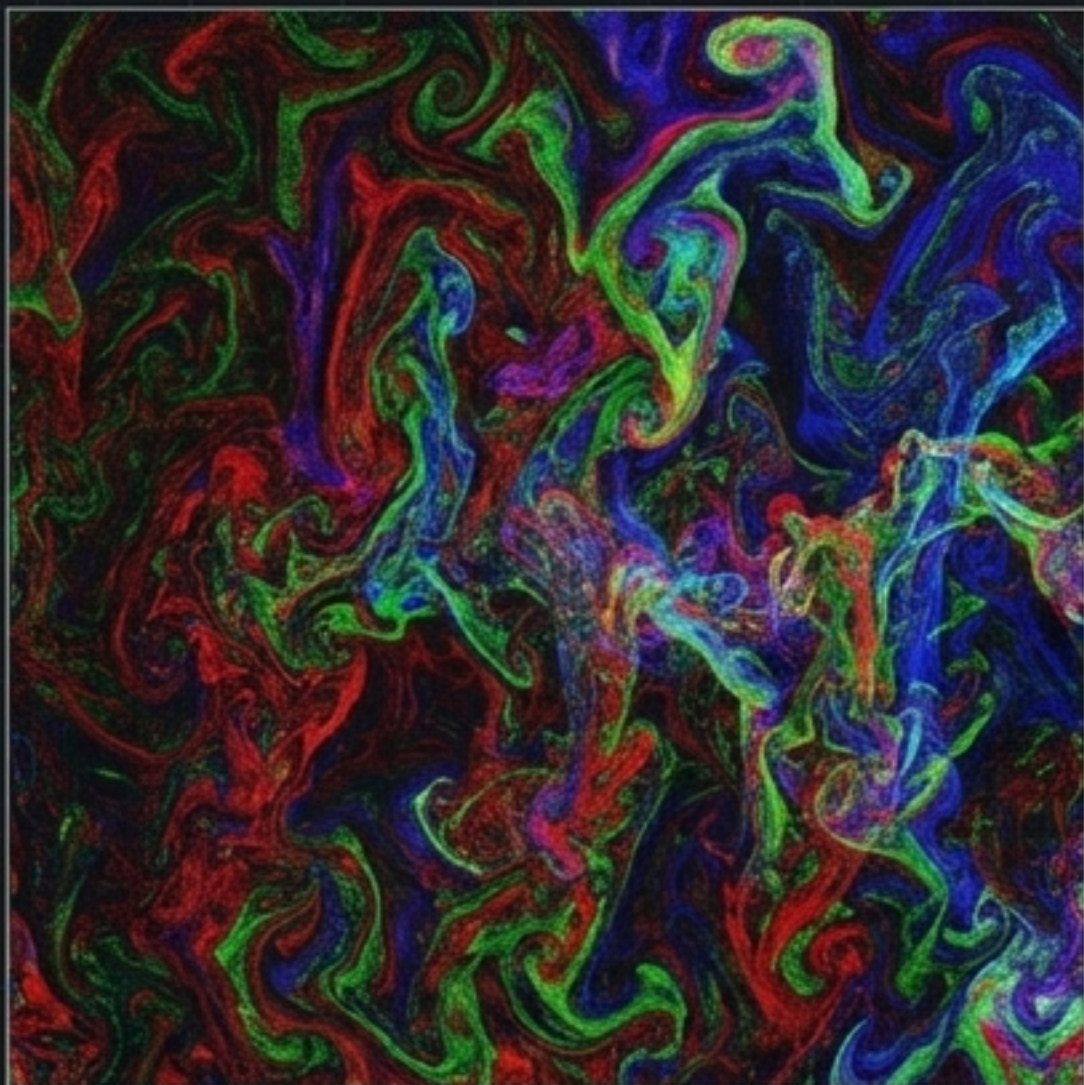
This map illuminates where the Ω equation spends the most computational time. It translates abstract mathematical gravity into thermal-visual weight, allowing immediate intuitive analysis of system stability.



Canvas 4: Combined RGB State-Space

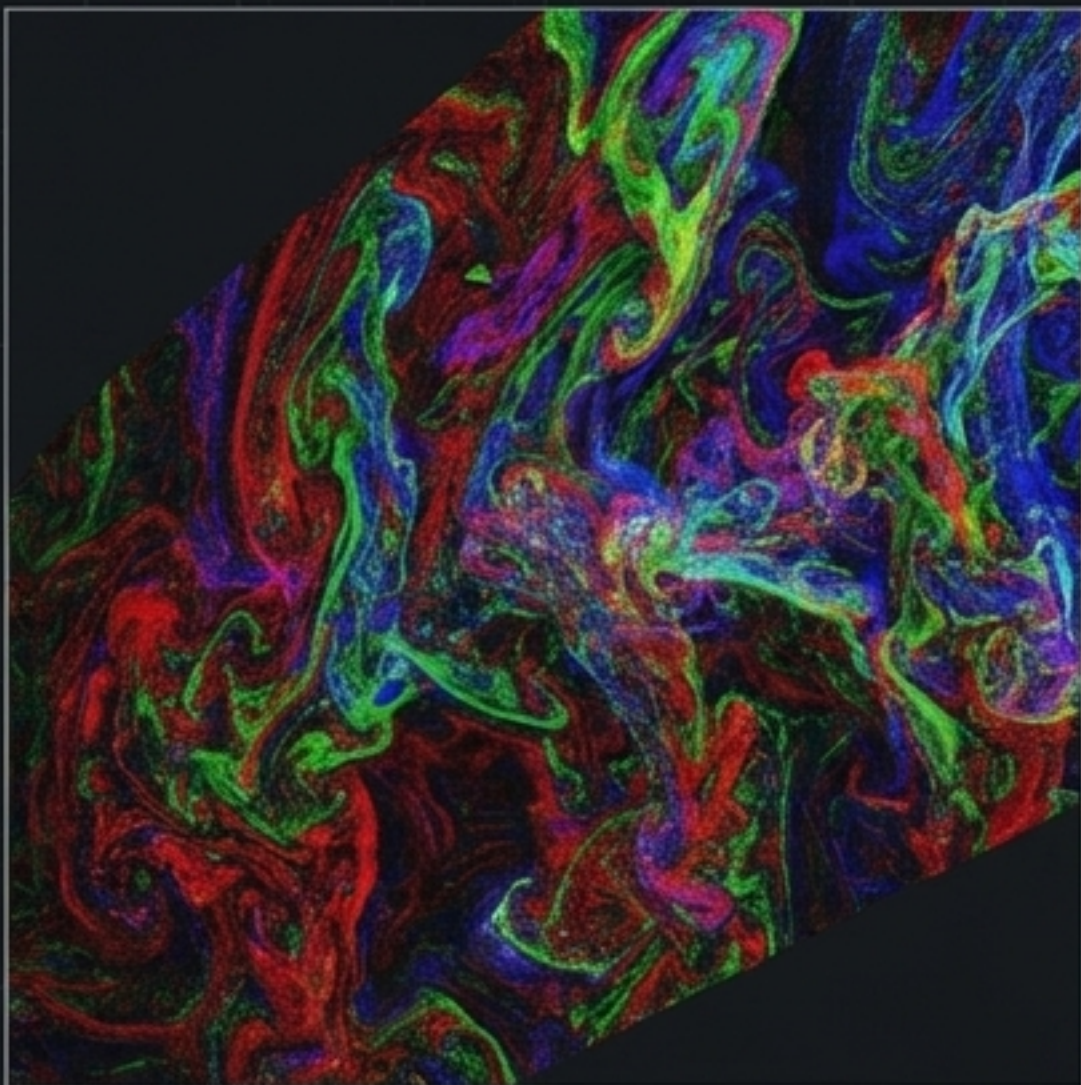
The synthesis of all dimensions. Different state variables are mapped directly to Red, Green, and Blue channels. Every single pixel of this multi-layered artifact is the direct, reproducible result of the Merkle-verified fossil ledger. True deterministic art.


```
python ophi_state_space_ai.py --initial 0.612 --bias -0.002 --alpha 1.001
```



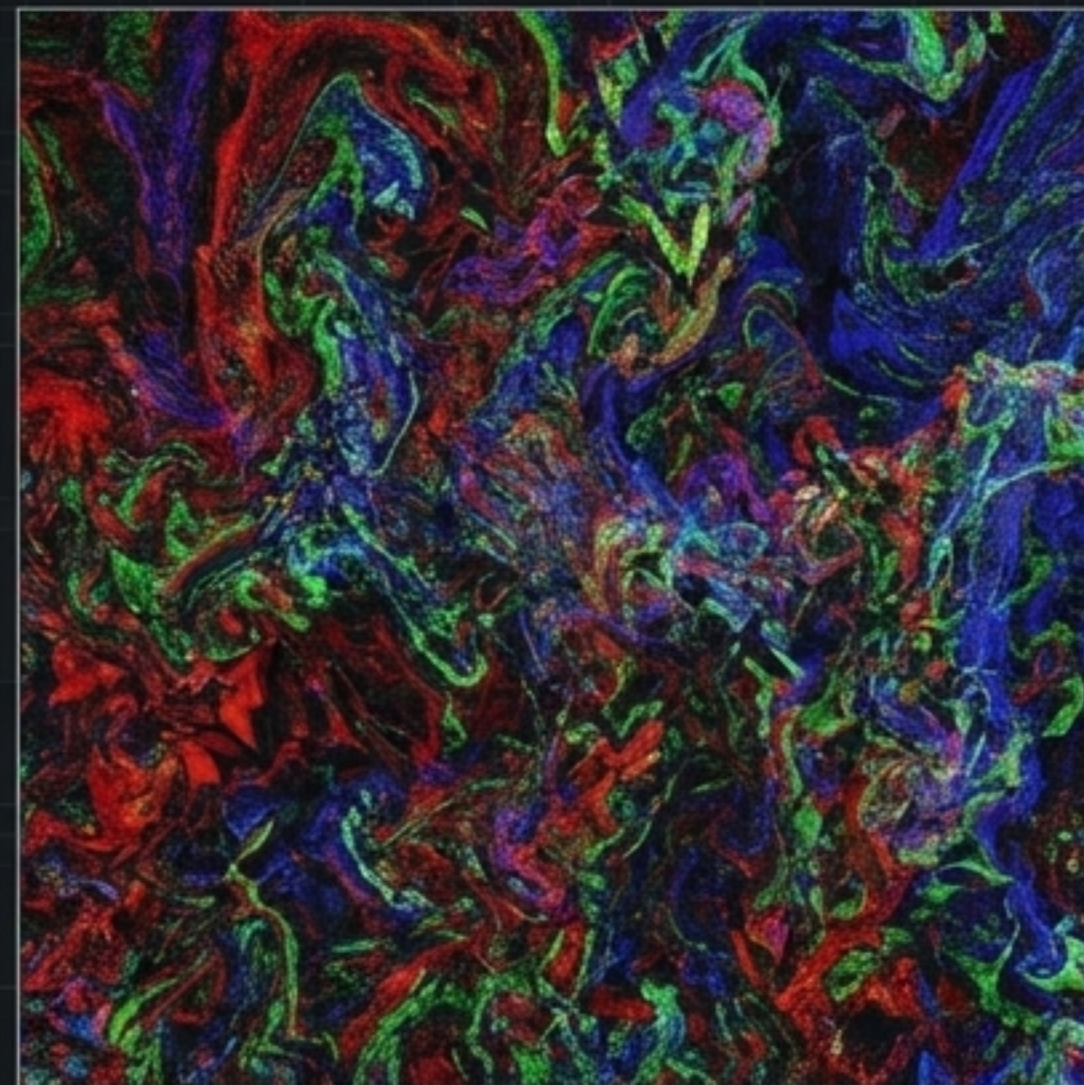
Base Rendering

Tied to --initial 0.612



Structural Shift

Tied to --bias -0.002



Evolution Variant

Tied to --alpha 1.001

Deterministic, yet infinite. Altering a single initial state by a thousandth of a decimal yields a wholly new, yet cryptographically locked, visual manifold.

